崑山科技大學日間部電子工程系四技課程科目表

Kun Shan University 4-Year College Curriculum (Day Division)

課程名稱(講授時數－實習時數－學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

114.04.08 系課程委員會修訂通過
114.04.22 院課程委員會議修正通過
114.07.22 校課程委員會通過
113 學年度入學適用
Applicable to the freshmen enrolled in Fall, 2024

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)							
校訂共同科目(合計 2 學分) General Courses (2 Credits)							
	生活實踐教育 (一) Living Practice (I) (0-1-1)	生活實踐教育 (二) Living Practice (II) (0-1-1)	服務學習與公民教育 Public Service and Civil Education (0-1-0)				
核心通識科目(合計 17 學分) Common Core Courses (17 Credits)							
體育（一） Physical Education (I) (2-0-2)	體育（二） Physical Education (I) (2-0-2)			生涯發展與職業倫理 Career Development and Work Ethics (工民商 2-0-2)	外語選讀 Foreign Languages ** (工商 2-0-2)		英文能力檢定* English Proficiency Examination* (1-0-1)
中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2)	中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2)						
英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2)						
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)							
分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)			
8-0-8	8-1-9	2-1-3	2-1-2	4-0-4 (工商創)	2-0-2 (工商創)		(1-0-1)

註：Note:

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。

'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。

'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

院訂基礎必修科目(合計 15 學分) **General Compulsory Courses of College(Total：18 Credits)**

機電整合專業模組課程（機械、電機、電子、機器人）

微積分(一) Calculus(I) (2-0-2)	微積分(二) Calculus(II) (2-0-2)	工程數學(一) Engineering Mathmatics(I) (3-0-3)			學術倫理* Academic Ethics (0-1-1)		
工程技術概論 Introduction to Engineering Technology (2-0-2)	程式設計 Programming (1-2-3)	科技英文 Technical English (2-0-2)					
4-0-4	3-2-5	5-0-5	0-0-0	0-0-0	0-1-1	0-0-0	0-0-0

註*:「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。The implementation of "Academic Ethics" and the acquisition of credits are handled accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.

院訂專業模組必修/必選科目(合計 21 學分) **Professional Modules Compulsory/Required Courses of College (Total：18 Credits)**

機電整合專業模組課程 (機械、電機、電子、機器人)

普通物理與實驗(一) General Physics and Experiment (I) (2-1-3)(必修) 可程式控制器實習 PLC practice (1-2-3)(必選)	機器人工程師實務 Robot Engineer Practice (1-2-3)(必選) 電路學 Electric Circuits (3-0-3)(必修)	電子學與實習 Electronics and Practice (1-2-3)(必修) 充電樁系統概論 Introduction to charging pile system （3-0-3）(必選)	車用晶片模組應用系統概論 Introduction to automotive chip module application system （3-0-3）(必選)				
3-3-6	4-2-6	4-2-6	3-0-3	0-0-0	0-0-0	0-0-0	0-0-0

專業必修科目(合計 38 學分) Required subjects for majors (Total：38 Credits)

電路板佈局設計與實習 Print Circuid Board Layout Design and Practice (2-1-3)	普通物理與實驗(二) General Physics and Experiment (II) (1-2-3) 數位邏輯設計與實習 Digital Logic Design and Practice (2-1-3)	基本電子學實習 Electronics Practice (0-2-2)	AIoT 基礎與實習 AIoT Fundamental and Practice (2-1-3) 進階電子學與實習 Advanced Electronics and practice (1-2-3)	實務專題(一) Practice and Study on Selected Topic (I) (0-1-1) 基礎量測及實習 Fundamental Measurement and Practice (1-2-3) 車用晶片模組規劃與設計 Automotive chip module planning and design （3-0-3） 車用晶片模組製造生產實務 Automotive chip module manufacturing and production practices （2-2-4）	實務專題(二) Practice and Study on Selected Topic (II) (0-1-1) 專業證照實習 Practical of Professional Certificate (0-1-1) 車用晶片模組測試檢驗實 Automotive chip module testing and inspection practices （2-2-4）	暑期校外實習 After School Practice (0-*-4)	
2-1-3	3-3-6	0-2-2	3-3-6	6-5-11	2-4-6	0-*-4	

專業選修科目，依每學期實際情況開課(選修學分 36 學分,本系至少 27 學分)

Major elective subjects are taught according to the actual conditions of each semester (36 credits for electives, at least 27 credits for this department)

電子工程基礎(一)** Fundamentals of Electronic Engineering (I) (0-3-3)	電子工程基礎(二)** Fundamentals of Electronic Engineering (I) (0-3-3)	電子工程進階(一)** Advance of Electronic Engineering(I) (0-3-3)	電子工程進階(二)** Advance of Electronic Engineering(II) (0-3-3)	電子工程應用** Application of Electronic Engineering (0-3-3)	機電整合實務** Mechatronics Practice (1-2-3) 智慧製造** Intelligent Manufacturing (3-0-3)	產線智慧化實務(一)** Practice on Intelligent Production Line (I) (1-2-3) 智慧生產管理** Intelligent Production Management (3-0-3)	產線智慧化實務(二)** Practice on Intelligent Production Line (II) (1-2-3)
互動多媒體程式設計 Interactive Multimedia Programming (3-0-3)		電化學* Electrochemistry (3-0-3) 電動車概論 Introduction to Electric Vehicle (3-0-3) 微處理機基礎 Basis of Microprocessor (1-2-3) 微算機系統設計實務 Practic on Micrcomputer System Design (2-1-3)	超大型積體電路設計與實習 Very Large Scale Integrated Circuit Design and Practice (2-1-3) 半導體元件與物理 Semiconductor Device and Physics (3-0-3) 數位系統設計與實習 Digital System Design and Practice (2-1-3) 儀表控制軟體與模擬 Instrument Controlling Software and Simulation (2-1-3) 儲能系統與產業概論 Introduction to Energy Storage Systems and Industry (3-0-3) 電動車機電整合概論 Introduction to Mechatronics of Electric Vehicles (3-0-3) 工程數學(二) Engineering Mathmatics(II) (3-0-3) 數位積體電路設計與佈局 Digital VLSI Design and Layout (2-1-3)	類比電路設計與應用 Analog Circuit Design and Application (3-0-3) 全客戶積體電路佈局設計與實習 Full Custom Integrated Circuit Design and Practice (2-1-3) 積體電路製程 Integrated Circuit Engineering (3-0-3) 通訊原理 Principle of Communications (3-0-3) 機器人智慧化實務與應用 Practice and Application of Robot Intelligence (3-0-3) 儲能系統應用與測試 Design and Application on Energy Storage System (3-0-3) 智慧感測與雲端控制應用 Intelligent Sensing and Cloud Control Applications (3-0-3) 電動車輛工程 Electric Vehicle Engineering (2-1-3) 感測器原理與應用實務** AI Sensor and It's Application (2-1-3) 充電樁規劃與設計 Charging pile planning and design (3-0-3) 充電樁製造生產實務 Charging pile manufacturing and production practices (2-2-4)	可程式控制器應用及實習 Application and Practice of PLC (2-1-3) 電力電子學與實習 Power Electronics and Practice (2-1-3) 智慧機器人控制系統 Intelligent Robot Control System (3-0-3) 自動控制 Automatic Control (3-0-3) 嵌入式系統設計與實習 Design and Practice of Embedded System (2-1-3) 智慧車用電子實務 Practice of intelligent Automotive Electronics (1-2-3) 車聯網原理與實作 Principles of Vehicle Intranet and the Laboratory (1-2-3) 車輛電機機械 Vehicle motor machinery (2-1-3) 智慧化科技應用實務 Practice on Intelligent Technology Application (3-0-3) 充電樁測試檢驗實務 Charging pile testing and inspection practices (2-2-4)	學期校外實習 Term School Practice (0-*-9) 學年校外實習(一) Annual Internship (I) (0-*-9) 汽機車電子技術 Vehicle Electronic Technology (3-0-3) 人機介面實務 Practice on Human- Machine Interface (3-0-3) 智慧車輛控制系統 Intelligent Vehicle Control Systems (2-1-3)	載具節能實務 Carrier Energy Saving Practice (3-0-3) 學期校外實習 Term School Practice (0-*-9) 學年校外實習(二) Annual Internship (II) (0-*-9) 光電半導體製程 Optoelectronics Semiconductor Processing Technolog (3-0-3) 燃料電池 Fuel Cells (3-0-3) 電動車檢診實務 Electric vehicle inspection practice (2-1-3)

最低畢業總學分為 130 學分 The minimum total credits for graduation is 130 credits

智慧運算與應用數位科技微學程(合計 11 學分)

	程式設計 Program Design (1-2-3)	通識(E) General Education(E) (2-0-2)	車用晶片模組應用系統概論 Introduction to automotive chip module application system （3-0-3）	車用晶片模組規劃與設計 Automotive chip module planning and design （3-0-3）				
	1-2-3	2-0-2	3-0-3	3-0-3				

學生須修畢上述本系智慧運算與應用數位科技微學程之課程始得畢業。

To be eligible for graduation, students must complete the department's Micro Program in Intelligent Computing and Applied Digital Technology.

註：Note:

1. 最低畢業學分為 130 學分，必修 103 學分、選修 27 學分。
   A minimum of 130 credits is required for graduation, including 110 required credits and 20 elective credits.
2. 課程分(一)(二)(三)(四)者，修課得不受前後段之限制。
   For courses labeled (I), (II), (III), or (IV), students may take them in any order without prerequisite restrictions.
3. 學生需至少修讀完成一跨院系學分學程，取得跨院系學分學程證書使得畢業。
   Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.
4. 學生須修畢本系智慧運算與應用數位科技微學程(通識教育 E、程式設計、車用晶片模組應用系統概論、車用晶片模組規劃與設計)，取得數位科技微學程證書使得畢業。
   Students must complete the Department's Micro Program in Intelligent Computing and Applied Digital Technology (including General Education E, Programming, Introduction to automotive chip module application system, and Automotive chip module planning and design) and obtain the Digital Technology Micro Program Certificate in order to graduate.
5. 學生須修畢本系電動車車用晶片模組應用技優學程（車用晶片模組應用系統概論、車用晶片模組規劃與設計、車用晶片模組製造生產實務、車用晶片模組測試檢驗實務)、充電樁開發製造技優學程(充電樁系統概論、充電樁規劃與設計、充電樁製造生產實務、充電樁測試檢驗實務)或本校跨領域學程，始得畢業。
   Students must complete one of the following programs to graduate：Advanced Technical Program in Automotive Chip Module Applications (Introduction to Automotive Chip Module Systems, Planning and Design, Manufacturing Practices, and Testing & Inspection), Advanced Technical Program in Charging Pile Development and Manufacturing (Introduction to Charging Pile Systems, Planning and Design, Manufacturing Practices, and Testing & Inspection), or an approved cross-college interdisciplinary program offered by the university.
6. 暑期校外實習之實施為**至少** 8 週 320 小時，另外訂定執行辦法。學期校外實習之實施為開設 9 學分以上，至少為期 4.5 個月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。若於四年級選修學期校外實習課程，則可抵免暑期校外實習及專業選修 7 學分。
   The summer off-campus internship must last at least 8 weeks (320 hours) and will follow separately defined implementation guidelines. The semester-long internship carries 9 or more credits and must last at least 4.5 months. During the internship, except for mandatory return-to-campus seminars or training sessions organized by the university, students are required to work full-time at the host organization. If a senior student enrolls in the semester-long internship course, it may be counted in place of the summer internship and 7 credits of professional electives.
7. 有**之科目為智慧電子學程科目，修畢 18 學分可取得學程證書。
   Courses marked with ** are part of the Smart Electronics Program. Students who complete 18 credits in these courses will be eligible to receive the Smart Electronics Program Certificate.